

Manjusha K

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Professional Summary

AI Engineer specializing in LLM applications, RAG pipelines, and multi-agent systems. Proficient in Python, LangChain, LlamaIndex, NLP, and cloud deployment (AWS). Delivered healthcare automation, fraud detection, and clinical monitoring platforms with measurable outcomes. Pursuing M.Sc. in Big Data Analytics focused on Generative AI.

Technical Skills

Languages & Frameworks: Python (Pandas, NumPy, scikit-learn, TensorFlow, Keras, PyTorch), R, SQL, Flask

AI / LLM Stack: LangChain, LlamaIndex, Flowise, RAG Pipelines, Prompt Engineering, LLM APIs (Gemini, Groq, Cohere, OpenAI), Vector Databases (Pinecone), AI Agents, Multi-agent Systems

MLOps & Deployment: Docker, FastAPI, AWS (EC2, S3, Terraform, AWS CLI), Streamlit, HuggingFace Spaces, n8n, Supabase, Git

ML & Deep Learning: Supervised/Unsupervised Learning, CNNs, ResNet50, Transfer Learning, Random Forest, SVM, Logistic Regression, ANOVA, Time Series Forecasting

Data & Analytics: Power BI, Tableau, Excel (Pivot Tables, Dashboards), MixPanel, A/B Testing, User Segmentation, KPI Tracking, Data Storytelling

NLP & Data Engineering: Natural Language Processing (NLP), Text Preprocessing, Data Pipelines, Document Parsing, OCR, ETL Workflows

Soft Skills: Analytical Thinking, Technical Communication, Team Leadership, Collaborative Project Execution

Experience

AI / GenAI Engineer Intern | *Trigent Software Pvt Ltd* Jan 2026 – Mar 2026 • Bengaluru

- Architected and shipped LLM-powered applications using Python, LangChain, Flowise, and n8n — including a medical report summarization system and a sentiment analysis engine serving clinical decision-making workflows.
- Deployed multi-agent AI systems (ChatGPT-style assistant, research agent, customer support bot) with API integrations via Flowise, reducing manual processing time for internal teams.
- Designed and deployed production RAG pipelines using document chunking, embedding generation, and Pinecone vector databases, enabling sub-second knowledge retrieval across enterprise document stores.
- Designed an ML-based fraud detection system using Random Forest and Logistic Regression with real-time Streamlit dashboards, achieving 95%+ accuracy in anomaly flagging on transaction data, significantly reducing false positives.
- Engineered healthcare AI workflows for medication safety, renal dose adjustment, and drug interaction detection using Supabase, n8n, and Streamlit — integrated into live clinical environments.
- Developed clinical monitoring platforms including chemotherapy toxicity surveillance and dialysis risk scoring, generating actionable alerts for healthcare staff.
- Spearheaded the development of an intelligent Rare Disease Diagnostic system using OCR/LLM-based clinical data extraction and HPO symptom mapping to predict rare disease classifications.
- Validated and debugged 800+ enterprise automation workflows, improving system reliability and cross-platform integration performance across production pipelines.

Projects

Handwriting-Based Parkinson's Disease Detection — *Tools: Python, TensorFlow, Keras, scikit-learn, ResNet50, SVM, Flask*

- Engineered a ResNet50 + SVM hybrid deep learning model to classify Parkinson's Disease from handwriting images, achieving 97.47% accuracy and AUC of 0.989 using stratified 5-fold cross-validation.

- Applied transfer learning and data augmentation to expand dataset from 3,500 to 7,100+ images, significantly improving model generalization and reducing overfitting.
- Shipped model as a Flask web application for real-time diagnostic inference.

EV Adoption & Air Quality Analysis — Electrifying Bengaluru — *Tools: Python, Pandas, NumPy, Matplotlib, scikit-learn, Excel*

- Analyzed 6 years (2018–2024) of EV adoption and air quality data to uncover emission trends and quantify the impact of government EV policy interventions.
- Developed forecasting models projecting future EV penetration and PM2.5 reduction under different policy scenarios, translating findings into infrastructure planning insights.

A/B Testing — E-Commerce Landing Page Conversion Optimization — *Tools: Python, Pandas, Statsmodels, Matplotlib, Excel*

- Designed and executed a statistically rigorous A/B test using a two-proportion z-test to measure new landing page impact on conversion rates.
- Delivered actionable insights that guided product team's data-driven redesign decision.

American Sign Language (ASL) Recognition Web App — *Tools: Python, TensorFlow, Keras, OpenCV, Flask, NumPy*

- Developed a CNN-based real-time ASL gesture recognition system, with data augmentation pipeline to reduce overfitting, deployed via Flask for browser-based interaction.

Cloud-Deployed Appointment Booking Web App — *Tools: Python, Flask, SQLAlchemy, SQLite, AWS EC2, HTML, Git*

- Engineered a full-stack Flask application with SQLAlchemy ORM, integrated payment gateway, and automated email delivery; deployed on AWS EC2 with production configuration.

Education

M.Sc. in Big Data Analytics | *St Joseph's University, Bengaluru* Jul 2024 – Present • SGPA: 8.06

B.Sc. in Statistics & Mathematics | *Mount Carmel College, Bengaluru* Aug 2021 – Apr 2024 • SGPA: 6.91

Certifications

- Google Analytics Certification — Great Learning
- Goldman Sachs Excel Skills for Business — Forage

Leadership & Activities

- Head of Media & Marketing Committee — Datagram 7.0, St. Joseph's University
- Core Team Member — Reel Run Association & Mathematics Association, Mount Carmel College
- Student Intern — Under25 Summit 2023